Regression Analysis 1

Lecture 10



Matrix Notation for Simple Regression Model :

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad , \; i = 1, 2, \ldots, n \text{ where } \varepsilon_i \sim N(0, \sigma^2)$$

$$\begin{aligned}
Y_1 &= \beta_0 + \beta_1 X_1 + \varepsilon_1 \rightarrow \text{individual 1} \\
Y_2 &= \beta_0 + \beta_1 X_2 + \varepsilon_2 \rightarrow \text{individual 2} \\
&\vdots \\
Y_n &= \beta_0 + \beta_1 X_n + \varepsilon_n \rightarrow \text{individuals } n
\end{aligned}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} \beta_0 + \beta_1 X_1 \\ \beta_0 + \beta_1 X_2 \\ \vdots \\ \beta_0 + \beta_1 X_n \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n \times 1}$$

$$\underset{n\times 1}{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}} = \underset{n\times 2}{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}} + \underset{n\times 1}{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}$$

$$Y_{n\times 1} = \underbrace{X_{n\times 2}}_{\text{design matrix}} \beta_{2\times 1} + \varepsilon_{n\times 1}$$

first column of X matrix is composed of 1's. (Because of intercept term(β_0) in the model)

$$Y_{n\times 1} = X_{n\times 2}\ \beta_{2\times 1} + \varepsilon_{n\times 1}$$

$$\varepsilon = Y - X\beta \quad \rightarrow \quad n\times 1$$

$$e = Y - X\hat{\beta} \quad \rightarrow \quad n\times 1 \quad : \text{residual vector}$$

$$S = \sum e_i^2 = e^T e = (Y - X\hat{\beta})^T (Y - X\beta) \rightarrow \min.$$

$$\frac{\partial S}{\partial \beta} = X^T Y - (X^T X)\hat{\beta} = 0$$

$$X^T Y = (X^T X)\hat{\beta}$$

$$(X^T X)^{-1} X^T Y = \hat{\beta}$$

$$X^T = X'$$

$$X'X = \underset{2\times n}{\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix}} \underset{n\times 2}{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}} = \underset{2\times 2}{\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix}}$$

$$X'y = \underset{2\times n}{\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix}} \underset{n\times 1}{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}} = \underset{2\times 1}{\begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}}$$

$$\sum Y_i = \hat{\beta}_0 n + \hat{\beta}_1 \sum X_i$$

$$\sum X_i Y_i = \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2$$

$$\downarrow \qquad \downarrow$$

$$X'Y = (X'X)\hat{\beta}$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$X'X = \begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} = n \begin{bmatrix} 1 & \bar{X} \\ \bar{X} & \frac{\sum X_i^2}{n} \end{bmatrix}$$

$$(X'X)^{-1} = \frac{1}{n\left(\frac{1}{n}\sum X_i^2 - \bar{X}^2\right)} \begin{bmatrix} \frac{1}{n}\sum X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{bmatrix}$$

$$= \frac{1}{\sum (X_i - \bar{X})^2} \begin{bmatrix} \frac{1}{n}\sum X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{bmatrix}$$

$$Var\{\hat{\beta}\} = \sigma^2 (X'X)^{-1}$$

$$\widehat{Var}\{\hat{\beta}\} = MSE\,(X'X)^{-1}$$

Multiple Regression Model :

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \ldots, n, \quad \varepsilon_i \sim N(0, \sigma^2)$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon \quad , \quad i = 1, 2, \ldots, n$$

k: number of independent variables

$p = k + 1$: number of parameters in the model.

$$Y_{n \times 1} = X_{n \times p} \, \beta_{p \times 1} + \varepsilon_{n \times 1}$$

$$\underset{p\times 1}{\hat{\beta}} = (X'X)^{-1} X'Y$$

$$\hat{\sigma}^2 = MSE = \frac{\sum e_i^2}{n-p} = \frac{e'e}{n-p}$$

$$Y = X\beta + \varepsilon$$
$$e = Y - X\hat{\beta}$$

$$S = e'e = (Y - X\hat{\beta})'(Y - X\hat{\beta})$$
$$= Y'Y - \underbrace{\hat{\beta}' X'Y}_{\text{scalar}} - \underbrace{Y'X\hat{\beta}}_{\text{scalar}} + \hat{\beta}'X'X\hat{\beta}$$
$$= Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\beta$$

$$\frac{\partial S}{\partial \hat{\beta}} = -2X'y + 2X'X\hat{\beta} = 0$$

$$(X'X)\hat{\beta} = X'y \rightarrow \text{normal equations}$$

$$(X'X)^{-1}(X'X)\hat{\beta} = (X'X)^{-1}X'y$$

$$\boxed{\hat{\beta} = (X'X)^{-1}X'y}$$

partial regression coefficient

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \cdots + \hat{\beta}_k X_k \quad : \text{fitted}$$

When the other independent variables in the model hold constant, one unit increase in X_1 variable, an average (mean) change in dependent variable (Y) is expected to be $\hat{\beta}_1$.

If the sign of $\hat{\beta}_1$ is positive → increase

" " " " " " negative → decrease

Predicted Values and Residuals:

$$\underset{n\times 1}{\hat{Y}} = \underset{n\times p}{X}\ \underset{p\times 1}{\hat{\beta}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}_{n\times 1}$$

: fitted (predicted) values of response variable

Hat matrix: (Şapka matrisi)

$$\hat{Y} = X\hat{\beta}$$

$$= \underbrace{X(X'X)^{-1}X'}_{H} Y = HY$$

where $\underset{n\times n}{H} = X(X'X)^{-1}X'$

H matrix is symmetric matrix $\rightarrow$ $H' = H$

H matrix is idempotent matrix $\rightarrow$ $H^2 = H$

$$\boxed{\hat{Y} = HY}$$

Residuals :

$$e = Y - \hat{Y}$$

$$= Y - HY = Y(I - H) \quad \text{where}$$

$(I-H)$ is also symmetric and idempotent matrix.

$$(I-H)' = I-H$$

$$(I-H)^2 = I-H$$

Variance - Covariance Matrix of Residuals :

$$\underset{n\times 1}{e} = (I-H)Y$$

$$Var\{e\} = (I-H)\,\sigma^2\,(I-H)'$$

$$= (I-H)(I-H)\,\sigma^2$$

$$= (I-H)^2\,\sigma^2$$

$$= (I-H)\,\sigma^2$$

$$\hat{\sigma}^2 = MSE = \frac{e'e}{n-p} = \frac{Y'Y - \hat{\beta}X'Y}{n-p}$$

$$\widehat{Var}\{e\} = MSE\,(I-H)$$

Preparing ANOVA Table :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ (Testing for overall significance of model)

H_1 : At least one $\beta_j \neq 0$ $(j = 1, \dots, k)$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2$$

$$= y'y - n\bar{y}^2 = y'y - \frac{(\sum y_i)^2}{n} \quad \text{where}$$

$$\frac{(\sum y_i)^2}{n} = \frac{1}{n} y' J y \quad \text{where } J \text{ } n \times n \text{ matrix with all elements } j = 1$$

$$SST = y'y - \frac{1}{n} y' J y$$

$$= y'y - n\bar{y}^2$$

$$SSE = \sum e_i^2 = e'e = (y - X\hat{\beta})'(y - X\hat{\beta})$$

$$= y'y - 2b'x'y + b'x'xb$$

$$= y'y - 2b'x'y + b' \underbrace{x'x (x'x)^{-1}}_{I} x'y$$

$$= y'y - 2b'x'y + b'x'y$$

$$= y'y - b'x'y \quad \text{where } \hat{\beta} = b$$

$$SSE = y'y - b'x'y$$

$$SST = y'y - n\bar{y}^2$$

$$SSR = SST - SSE = b'x'y - n\bar{y}^2$$

$Var\{b\} = ?$ $\quad b = \hat{\beta}_{p\times 1}$

$$b = \underbrace{(x'x)^{-1}x'}_{\text{scalar matrix } A}y$$

$$Var\{b\} = Var(Ay) = A\,Var(y)\,A' \quad \text{where} \quad \sigma^2\{y\} = \sigma^2 \cdot I$$

$$\sigma^2\{y\} = \sigma^2 \cdot I = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix}$$

$Cov(\varepsilon_i, \varepsilon_j) = 0 \quad (i \neq j)$

homoscedasticity $(i = j)$

$(x'x)$ and $(x'x)^{-1}$ are symmetric matrices.

$$A' = x\,(x'x)^{-1}$$

$$Var\{b\} = (x'x)^{-1}x'\,\sigma^2 I\,x\,(x'x)^{-1}$$

$$= \sigma^2\,(x'x)^{-1}\underbrace{x'x\,(x'x)^{-1}}_{I}$$

$$= \sigma^2\,(x'x)^{-1}$$

$$\boxed{Var\{b\} = MSE\,(x'x)^{-1}}$$

Mean Response : (for β_0)

$\hat{Y}_h = X_h b$ where $X_h = [1 \;\; X_h]'$

$Var(\hat{Y}_h) = X_h' \, Var\{b\} \, X_h$

$= X_h' \, \sigma^2 (X'X)^{-1} X_h$

$= \sigma^2 \, X_h' (X'X)^{-1} X_h$

$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$

$X_1 = 1 \;, X_2 = 2 \;,$

$\hat{Y} = [1 \;\; 1 \;\; 2] \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$ where $X_h = [1 \;\; 1 \;\; 2]$

($X_1 = 1$, $X_2 = 2$)

$\widehat{Var}(\hat{Y}_h) = MSE \, X_h' (X'X)^{-1} X_h \rightarrow \sqrt{\widehat{Var}} = se(\hat{Y})$

Variance formula for individual (or new) observation :

$\hat{Y}_0 = X_0 b$ where $X_0 = [1 \;\; X_0]'$

$Var(Y_0) = \sigma^2 X_0' (X'X)^{-1} X_0 + \sigma^2 I$

$= \sigma^2 (1 + X_0' (X'X)^{-1} X_0)$

$\widehat{Var}(Y_0) = MSE (1 + X_0' (X'X)^{-1} X_0)$

$\sqrt{\widehat{Var}(Y_0)} = se(Y_0)$

ANOVA Table :

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$

$H_1 :$ At least one $\beta_j \neq 0 \quad (j = 1, \ldots, k)$

Source of Variation	SS	df	MS	F
SSR	$SSR = b'X'y - n\bar{y}^2$	k	$MSR = SSR/k$	$F = \frac{MSR}{MSE}$
SSE	$SSE = y'y - b'X'y$	$n-k-1 = n-p$ where $p = k+1$	$MSE = SSE/n-p$	
SST	$SST = y'y - n\bar{y}^2$	$n-1$		

$$F = \frac{MSR}{MSE} \sim F_{(\alpha, k, n-p)} \rightarrow \text{significance level}$$

If $F \leq F_{(\alpha, k, n-p)}$ → H_0 is not rejected

$F >$ " → H_0 is rejected. Model is significant.

(At least one $\beta_j \neq 0$)

(Multiple) Determination Coefficient

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \quad 0 \le R^2 \le 1$$

$+\sqrt{R^2} = +R$: Multiple correlation coefficient

$$R^2 = \frac{b'x'y - n\bar{y}^2}{y'y - n\bar{y}^2} = 1 - \frac{y'y - b'x'y}{y'y - n\bar{y}^2}$$

Adjusted R^2 : R_a^2

$$R_a^2 = 1 - \frac{\frac{SSE}{n-p}}{\frac{SST}{n-1}} = 1 - \left(\frac{n-1}{n-p}\right)\left(\frac{SSE}{SST}\right)$$

Significance Test for Coefficient:

$H_0: \beta_k = 0 \qquad (k = 0, \ldots, k)$

$H_1: \beta_k \neq 0$
$<$
$>$

$$t = \frac{b_k}{se(b_k)} \sim t_{(n-p)}$$

If $|t| > t_{(1-\frac{\alpha}{2}, n-p)}$ H_0 is rejected

$\neq$

CI for Coefficient

$$b_k \pm t_{(1-\frac{\alpha}{2}, n-p)} \; se(b_k)$$

where $se(b_k)$ is the square root of diagonal elements of $\widehat{Var}(b)$ matrix

$$\widehat{Var}(b) = MSE\,(X'X)^{-1}$$

$$\widehat{Var}(b) = \begin{bmatrix} \widehat{Var}(b_0) & \widehat{Cov}(b_0, b_1) & \cdots & \widehat{Cov}(b_0, b_k) \\ & \widehat{Var}(b_1) & \cdots & \widehat{Cov}(b_1, b_k) \\ \text{Sym.} & & \ddots & \vdots \\ & & \cdots & \widehat{Var}(b_k) \end{bmatrix}$$

for $k=0 \rightarrow (1,1)$ element of this matrix $\rightarrow \hat{\beta}_0$

for $k=1 \rightarrow (2,2)$ " " " " $\rightarrow \hat{\beta}_1$

for $k=k \rightarrow (k+1, k+1)$ " " " $\rightarrow \hat{\beta}_k$

$$se(b_k) = \sqrt{\widehat{Var}(b_k)}$$

Confidence Interval (CI) for mean response and individual (or new) observation : prediction

$(1-\alpha)\%$ CI for mean response :

$$\hat{y}_h \pm t_{(1-\frac{\alpha}{2},\, n-p)}\, se(\hat{y}_h)$$

$(1-\alpha)\%$ CI for individual observation or prediction

$$\hat{y}_0 \pm t_{(1-\frac{\alpha}{2},\, n-p)}\, se(y_0)$$

where $se(y_0) = \sqrt{MSE(1 + x_0'(x'x)^{-1}x_0)}$ and $x_0 = \begin{bmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0k} \end{bmatrix}$

ex : $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$, $n = 11$

i	x_1	x_2	y	x_1^2	x_2^2	$x_1 y$	$x_2 y$	y^2
1	1	8	6					
2	4	2	8					
3	9	−8	1					
4	11	−10	0					
5	3	6	5					
6	8	−6	3					
7	5	0	2					
8	10	−12	−4					
9	2	4	10					
10	7	−2	−3					
11	6	−4	5					
	66	−22	33				85	142

$$X = \begin{bmatrix} 1 & 1 & 8 \\ 1 & 4 & 2 \\ 1 & 9 & -8 \\ 1 & 11 & -10 \\ 1 & 3 & 6 \\ 1 & 8 & -6 \\ 1 & 5 & 0 \\ 1 & 10 & -12 \\ 1 & 2 & 4 \\ 1 & 7 & -2 \\ 1 & 6 & -4 \end{bmatrix}_{11 \times 3} \qquad y = \begin{bmatrix} 6 \\ 8 \\ 1 \\ 0 \\ 5 \\ 3 \\ 2 \\ -4 \\ 10 \\ -3 \\ 5 \end{bmatrix}_{11 \times 1}$$

$$X'X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 4 & \dots & 6 \\ 8 & 2 & \dots & -4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 8 \\ 1 & 4 & 2 \\ \vdots & \vdots & \vdots \\ 1 & 6 & -4 \end{bmatrix} = \begin{bmatrix} 11 & 66 & -22 \\ 66 & \Sigma x_1^2 & \Sigma x_1 x_2 \\ \Sigma x_2 & \Sigma x_1 x_2 & \Sigma x_2^2 \end{bmatrix}$$

$$(X'X)^{-1} = \frac{1}{\det(X'X)} \, Adj(X'X)$$

$$(X'X)^{-1} = \begin{bmatrix} 4.3705 & -0,8495 & -0,4386 \\ -0,8495 & 0,1690 & 0,0822 \\ -0,4386 & 0,0822 & 0,0422 \end{bmatrix}$$

$$X'y = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 4 & \dots & 6 \\ 8 & 2 & \dots & 4 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 8 \\ \vdots \\ 5 \end{bmatrix} = \begin{bmatrix} \Sigma y \\ \Sigma x_1 y \\ \Sigma x_2 y \end{bmatrix} = \begin{bmatrix} 33 \\ 85 \\ 142 \end{bmatrix}$$

$(X'X)\hat{\beta} = X'y \rightarrow$ normal denklemler

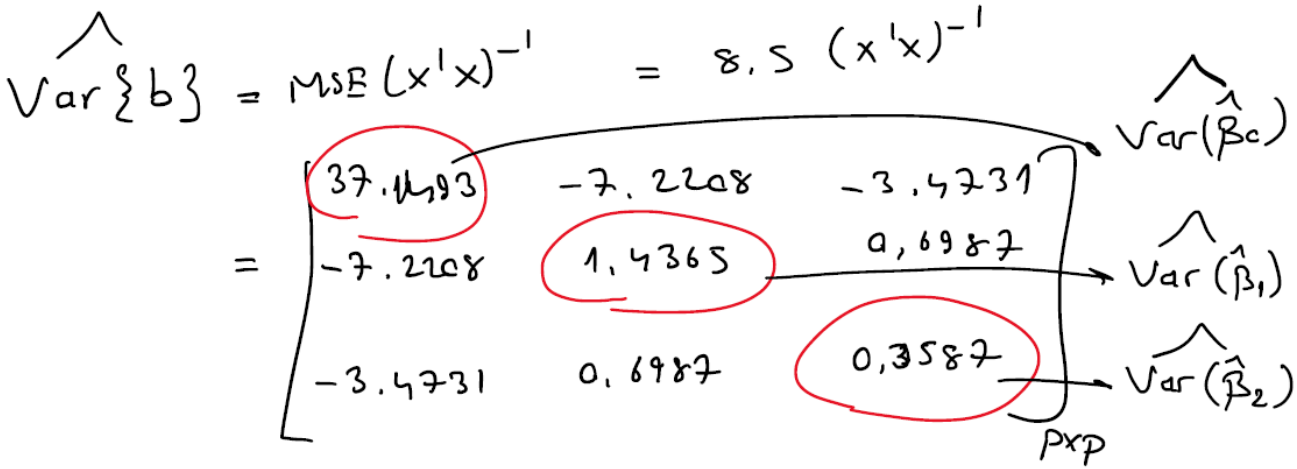
a) $\hat{\beta} = b = (X'X)^{-1}X'y$

$= (X'X)^{-1}\begin{bmatrix} 33 \\ 85 \\ 142 \end{bmatrix} = \begin{bmatrix} 14 \\ -2 \\ -0.5 \end{bmatrix} = b$

$\hat{y} = 14 - 2x_1 - 0.5x_2$ fitted (estimated) model

b) $\widehat{Var}(b) = ?$

$\hat{\sigma}^2 = MSE = \frac{e'e}{n-p} = \frac{y'y - b'X'y}{n-p} = \frac{68}{8} = 8,5$


$\widehat{Var}\{b\} = MSE(X'X)^{-1} = 8.5\,(X'X)^{-1}$
$= \begin{bmatrix} 37.4493 & -7.2208 & -3.4731 \\ -7.2208 & 1.4365 & 0.6987 \\ -3.4731 & 0.6987 & 0.3587 \end{bmatrix}$
$\widehat{Var}(\hat{\beta}_c)$
$\widehat{Var}(\hat{\beta}_1)$
$\widehat{Var}(\hat{\beta}_2)$
pxp

$\widehat{Var}(\hat{\beta}_0) = 37.1493 \rightarrow se(\hat{\beta}_0) = \sqrt{37.1493}$

$\widehat{Var}(\hat{\beta}_1) = 1.4365 \rightarrow se(\hat{\beta}_1) = \sqrt{1.4365}$

$\widehat{Var}(\hat{\beta}_2) = 0.3587 \rightarrow se(\hat{\beta}_2) = \sqrt{0.3587}$

c) $R^2 = ?$

$$R^2 = \frac{SSR}{SST} = \frac{b'X'y - n\bar{y}^2}{y'y - n\bar{y}^2} = \frac{122}{190} = 0,642$$

64,2% of total variation of Y can be explained by the model.

$R^2_a = ?$

d) Is the model significant ? $\alpha = 0.05$

$H_0: \beta_1 = \beta_2 = 0$

$H_1: \beta_j \neq 0 \quad (j = 1, 2)$

$F_{(0.05\ ;\ 2\ ,\ 8)} = 4.46$

Source	SS	df	MS	F
SSR	122	2	61	7.18
SSE	68	8	8,5	
SST	190	10		

$7.18 > 4.46$. H_0 is rejected model is significant.

e) $H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

$$t = \frac{-2}{\sqrt{1.4365}} \quad , |t| < t_c$$

$t_{(1-\frac{\alpha}{2}, 8)} = t_c > |t| \rightarrow H_0$ is not rejected.

β_1 is insignificant. Or,

The contribution of X_1 to the model is insignificant.

f) $X_1 = 3$ and $X_2 = 5$.

Find 95% CI for prediction (individual) or

Find 95% Prediction Interval.

$\hat{y}_0 = 14 - 2(3) - 0,5(5) = 5.5$ or $\hat{y}_0 = X_0' b$

$$X_0' (X'X)^{-1} X_0 = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix} (X'X)^{-1} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = 0,2225$$

$$5.5 \pm 2.306 \sqrt{8,5\,(1 + 0.2295)}$$

$$\downarrow$$

$$t_{(1-\frac{\alpha}{2},\,8)}$$

$$-1,9548 \leq y_0 \leq 12.9548$$

9) $X_1 = 3$ and $X_2 = 5$

Find 95% CI for mean response at X_h.

$$X_h' = [1 \quad 3 \quad 5]$$

$$\hat{\sigma}^2 = MSE = 8,5$$

$$5.5 \pm 2.306 \sqrt{8,5\,(0.2295)}$$

$$\hat{y} = 14 - 2X_1 - 0.5X_2$$

- 14: intercept (sabit terim)
- $-2X_1$: slope (eğim)
- $-0.5X_2$: slope

$\hat{\beta}_1$: When the X_2 holds constant, with 1 unit increase in X_1, on average causes 2 unit decrease in y variable.

ex: $y, x_1, x_2, x_3, n = 8$

i	x_{i1}	x_{i2}	x_{i3}	y
1	3	4	10	2
2	4	7	8	3
3	4	3	9	4
4	6	8	8	6
5	7	10	7	8
6	10	11	6	9
7	11	12	5	11
8	12	10	3	13

a) $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$

$$X = \begin{bmatrix} 1 & 3 & 4 & 10 \\ 1 & 4 & 7 & 8 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 12 & 10 & 3 \end{bmatrix}_{8\times 4} \qquad y = \begin{bmatrix} 2 \\ 3 \\ \vdots \\ 13 \end{bmatrix}_{8\times 1}$$

$$X'X = \begin{bmatrix} 8 & 57 & 65 & 56 \\ 57 & 491 & 532 & 346 \\ 65 & 532 & 603 & 413 \\ 56 & 346 & 413 & 428 \end{bmatrix}$$

$$X'y = \begin{bmatrix} 56 \\ 493 \\ 530 \\ 332 \end{bmatrix}$$

$$(X'X)^{-1} = \frac{1}{36745} \begin{bmatrix} 1690317 & -96853 & 3222 & -14575 \\ -96853 & 7452 & -2028 & 865 \\ 3222 & -2028 & 1972 & -685 \\ 14597S & 8605 & -685 & 12890 \end{bmatrix}$$

$$\hat{\beta} = (X'X)^{-1} X'Y = b = \begin{bmatrix} 4.169 \\ 0.873 \\ -0.044 \\ -0.433 \end{bmatrix} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix}$$

$$\hat{Y} = 4.169 + 0.873\, X_1 - 0.044 X_2 - 0.433\, X_3$$

b) $n = 8$

$$\bar{Y} = \frac{56}{8} = 7$$

$$Y'Y = \sum Y_i^2 = 500$$

$$b'X'Y = 496.777$$

$$SST = Y'Y - n\bar{Y}^2 = 500 - 8(7^2) = 108$$

$$SSR = b'X'Y - n\bar{Y}^2$$

$$= 496.777 - 8(7^2) = 104.777$$

$$SSE = y'y - b'x'y = 500 - 496.777 = 3.223$$

$$R^2 = \frac{104.777}{108} = 0,970 \rightarrow 92\%$$

$$R^2_{adj} = 1 - \frac{3.223 / (8-4)}{108 / 7} = 0,947 \rightarrow 94.7\%$$

$$\widehat{V(b_1)} = 0.8058\left(\frac{7452}{36745}\right) = 0.1634$$

$$se(b_1) = \sqrt{0.1634} = 0.404$$

$$t_{0.025;\,4} = 2.776$$

$$0.873 - 2.776(0.404) < \beta_1 < 0.873 + 2.776(0.404)$$

$$-0.249 < \beta_1 < 1.995$$

$\downarrow$

CI includes zero. From here

$H_0: \beta_1 = 0$ is not rejected (if hypothesis test was conducted)

Is the model significant ?

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$

$H_1 :$ At least one $\beta_j \neq 0 \quad (j = 1, 2, 3)$

$SSR = 104.727$

$SSE = 3.223$

$MSR = \frac{104.727}{3} = 34.925$

$MSE = \frac{3.223}{8-4} = 0,8050$

$F = \frac{MSR}{MSE} = \frac{34.925}{0,8050} = 43.342$ or

$F = \frac{R^2 / (p-1)}{(1-R^2) / (n-p)} = \frac{0,97 / (4-1)}{(1-0,97) / (8-4)} = 43.11$

$\alpha = 0.01 \quad , F_{(\alpha, 3, 4)} = 16.69 = F_c$

$F > F_c$ so

H_0 is not rejected. At least one $\beta_j \neq 0$.

$H_0 : \beta_3 = 0$

$H_1 : \beta_3 \neq 0$

$b_3 = -0,433$

$\sqrt{\widehat{Var}(b_3)} = 0,531 \doteq se(b_3)$

$$t = \frac{-0,433 - 0}{0,531} = -0,815$$

$|t| < t_c = t_{(1-\frac{\alpha}{2}, 4)} \rightarrow H_0$ is not rejected. β_3 is insignificant.